Exploring (psycho)linguistically-informed approaches to ASR dataset development for minoritized language varieties

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Introduction

Accent bias in automatic speech recognition (ASR) systems

- Automatic speech recognition (ASR) systems have become ubiquitous, and provide an increasingly unavoidable means for human-computer interaction
- ASR systems typically perform better for speakers of:
  - Historically favored languages (Sprugnoli et al., 2017) and accents (Markl 2022)
  - So-called "native" (as opposed to "non-native") accents (DiChristofano et al. 2023)
- ASR models are built from transcribed speech corpora – the so-called "ground truth"
  - Corpus quality depends on large and representative corpora with accurate transcriptions
- Solutions to ASR system bias typically call for larger datasets, assuming that more data from diverse speakers will reduce accent bias (Harwell 2018)
- However, more data alone cannot solve the problem of disparate labeling errors for some accents (Prinos et al. 2024)
- ***The goal of our ongoing project is to mitigate the biases that permeate ASR by exploring psycholinguistically-informed methods for dataset construction***

Insights from psycholinguistics

Listener factors

- Similarities between the linguistic sound systems of speakers and listeners can support listening comprehension accuracy and speed
- Adank et al. (2009) found a listening comprehension advantage for speakers of so-called Standard British English listening to other speakers with their same accent compared to Glaswegian English-accent speech
- Listeners whose first language is Mandarin outperformed those whose first language is English when listening to Mandarin-accented English speech (the Interlanguage Speech Intelligibility Benefit; Hayes-Harb et al. 2008)
- Patwari and colleagues independently hypothesized the existence of an intelligibility benefit, recommending that researchers "ensure that transcribers are fluent in the accent of the speakers whose speech they transcribe" (Prinos et al. 2024: 1251)

Task factors

- Listeners can rapidly adapt to unfamiliar accents (Clarke & Garrett, 2004)
- Transcriptions are more accurate in single-talker conditions (where speech samples are blocked by speaker) than in mixed-talker conditions (where speech samples from multiple talkers are mixed; Mullennix et al. 1989)
- Especially when the accent is unfamiliar to the transcribers and/or the signal is degraded by noise (Tamati & Pisoni 2014)

Is transcription accuracy higher when:
RQ1 - talkers and listeners have similar linguistic backgrounds?
RQ2 - listeners self-identify as familiar with the talker's accent?
RQ3 - audio recordings are presented in single- versus mixed-talker blocks?

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Methods

Participants

- Recruited 258 participants via Prolific.co (~$12/hour, selected only for self-reported English language fluency)
- Randomly assigned to one of 4 conditions: 3 single-talker conditions and one multi-talker condition

Talker Condition	Talker ID	Transcriber First Language	N	Mean Age	Female / Male / Prefer to self-describe
Single Talker	Mandarin 43	English	29	37	19/10/0
		Mandarin	18	30	8/10/00
		Other	23	33	12/11/00
	Mandarin 39	English	42	33	24/18/0
		Mandarin	8	33	5/3/00
		Other	12	31	6/6/00
	Mandarin 35	English	34	35	24/10/0
		Mandarin	15	34	12/2/01
		Other	14	37	6/8/00
Multiple Talkers	Mandarin 35	English	34	43	19/15/0
	Mandarin 39	Mandarin	12	34	8/4/00
	Mandarin 43	Other	17	34	8/9/00

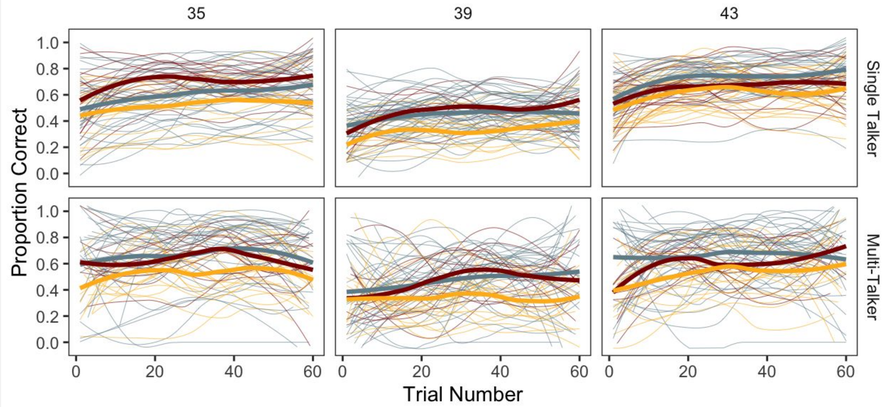
Stimuli

- 60 sentences produced by three Mandarin talkers (35, 39, 43) from the ALLSTAR corpus (Bradlow n.d.)
- *num_tlkrs*: Single (all 60 sentences by a single talker), Multi (20 randomly-selected sentences by each of the 3 talkers)

Procedures

- Implemented using PsychToolkit (Stoet 2010, 2017)
- Pre-task questionnaire
  - ***first_lang***: Do you speak Mandarin (as L1, as L2, no)? Do you speak English (as L1, as L2)? Coded as Transcriber L1 = English, Mandarin, Other
  - ***familiarity_mandarin***: Use the (0-10) slider to indicate how familiar you are with Mandarin-accented English
  - ***beliefs_composite***: The goal of a language learner should be to pronounce the language like a native speaker; The goal of a language learner should be to speak with a standard accent in the language; Non-native accents make it hard for me to understand speech; Non-standard accents make it hard for me to understand speech (0-10 slider; composite score computed via factor analysis using the psych package in R)
  - ***good_at_accents***: I am good at understanding accents that are different from mine (0-10 slider)
  - ***interact_accents***: I interact frequently with people whose accents are different from mine (0-10 slider)
- Sentence transcription task
  - Sentence transcription accuracy was computed using autoscore package in R (Barrett et al. 2019)
- Post-task questionnaire
  - ***post_tlkrlang_coded***: To the best of your ability, please describe the dialects/accents of the speakers you just listened to (for example, by country, region, language background); coded for containing "Chinese" or "Mandarin"
  - ***post_accent_fam***: How familiar are you with the accent of the speaker you just listened to? (0-10 slider)

Results



Transcription accuracy (Proportion Correct) over time (Trial Number) for each participant (thin lines) and first language group (thick lines); colors indicate first language. Facets indicate talker condition (columns: talker_id; rows: number of talkers (single, multi).

Effect	Sum Sq	Mean Sq	NumDF	DenDF	F value	P value	Note
order	0.016	0.016	1	15164.95	0.461	0.497	
order_sq	**2.368**	**2.368**	**1**	**15165.498**	**69.246**	**<.001**	**cubic model has best fit; listeners adapted over the course of the task**
order_cu	**1.028**	**1.028**	**1**	**15165.288**	**30.068**	**<.001**	
first_lang	**1.006**	**0.503**	**2**	**257.444**	**14.715**	**<.001**	**RQ1 - English, Mandarin, Other**
familiarity_mandarin	0.105	0.105	1	257.444	3.061	0.081	RQ2 - pre-task self-assessment of familiarity with Mandarin-accented English
num_tlkrs	0.084	0.084	1	257.444	2.466	0.118	RQ3 - Single (one), Multi (three)
talker_id	**10.01**	**5.005**	**2**	**363.79**	**146.377**	**<.001**	**Mandarin35, Mandarin39, Mandarin43**
beliefs_composite	0.009	0.009	1	257.444	0.266	0.607	pre-task language and accent beliefs
good_at_accents	0.008	0.008	1	257.444	0.227	0.634	pre-task self-assessment of facility with unfamiliar accents
post_tlkrlang_coded	**0.66**	**0.66**	**1**	**257.444**	**19.291**	**<.001**	**post-task open-ended identification of the accent just heard**
interact_accents	0.012	0.012	1	257.444	0.356	0.552	pre-task self-assessment of amount of experience with unfamiliar accents
post_accent_fam	0.017	0.017	1	257.444	0.498	0.481	post-task self-assessment of amount of experience with the accent just heard
num_tlkrs:talker_id	**0.413**	**0.207**	**2**	**363.79**	**6.043**	**0.003**	**two-way interaction of number of talkers and talker ID**

Summary of Findings

- RQ1 - **No.** English L1 advantage over non-Mandarin transcribers; no L1 Mandarin advantage
- RQ2 - **Maybe.** Marginal self-reported familiarity with Mandarin-accented English advantage
- RQ3 - **No.** No main effect of single vs. multi-talker; interaction of number of talkers with talker_id → depends on who is talking
- Participants adapted over the course of the 60-sentence task; no additional improvement after trial 40
- Main effect of talker_id (in addition to interaction of talker_id and num_tlkrs)
- Post-task identification of the talker as having a Chinese or Mandarin language background was significant
- Pre-task self-reported beliefs about language and accents and facility/xperience with accents), and post-task self-assessment of familiarity with the accent just heard were not significant

Next Steps & Implications

- ***Our ultimate goal***: mitigate the biases that permeate ASR ***while continuing to problematize ASR systems***
  - Consent and compensation of talkers; privacy, community language values and practices
  - Consent and compensation of transcribers; harmful content, worker exploitation
  - "Native speakerhood" and "standard language" ideologies
  - Essentializing language users: monolingual, multilingual, dialect, and interactions with race, ethnicity, nationality, etc.
- ***Our next step***: Examine the generalizability of these listener and task factors under conditions that more closely resemble crowdsourced ASR dataset development
  - Repeat these experiments with new materials: Sentences from Mozilla CommonVoice produced by consenting and compensated speakers of Bangladeshi and Nigerian English

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